Heejung Choi

Researcher, Asan Institute for Life Sciences, Asan Medical Center
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Education

University of Ulsan College of Medicine — Seoul, South Korea
M.S., Medical Engineering — Sep 2021 – Aug 2023

Seoul Women's University — Seoul, South Korea
B.S., Biology & Bioinformatics — Mar 2017 – Feb 2021

Research Experience

Asan Medical Center — Seoul, South Korea
Researcher, Department of Information Medicine — Mar 2021 – Present

- Designed deep-learning and machine-learning models for clinical prediction tasks using electronic health records.
- Applied explainable-AI techniques (SHAP, attention visualization) for model interpretation.
- Conducted a clinician reader study to compare model predictions with clinical judgment.
- Fine-tuned open-weight instruction LLMs using 4-bit quantization and LoRA on a GPU (RTX 3090, 24GB).
- Transformed raw electronic health records into model-specific representations.
- Drafted an IRB protocol for approval of human data access for biomedical research.
- Completed Good Clinical Practice (GCP) training.
- Contributed to writing a research grant proposal.

Sungkyunkwan University — Suwon, South Korea
Undergraduate Research Intern, Department of Molecular Cell Biology — Jan 2020 – Sep 2020

- Performed molecular biology experiments, including Western blotting, DNA preparation, and cloning
- Analyzed and visualized NCBI GEO transcriptomic datasets using R

Honors & Awards

Study Abroad Scholarship, $80,000
National Institute for International Education (NIIED) — 2025

- Waived after non-admission for the associated 2026 admission cycle

Bio-signal & EMR AI Specialist Training Program — 1st Prize (among 6 teams)
Korea Human Resource Development Institute for Health and Welfare (KOHI) — 2022

- Developed and validated an emergency department disposition prediction model using the MIMIC-IV database

Outstanding Talent Scholarship
Seoul Scholarship Foundation — 2020

Publications

First-author

- **Choi H**, Kang HJ, Ahn I, Gwon H, Kim Y, Seo H, Cho HN, Han J, Kim M, Kee G, Park S, Kwon O, Roh JH, Kim AR, Kim JH, Jun TJ, Kim YH. Machine learning models to predict the warfarin discharge dosage using clinical information of inpatients from South Korea. *Scientific Reports*. 2023 Dec 18;13(1):22461.

- **Choi H**, Kim Y, Kang H, Seo H, Kim M, Han J, Kee G, Park S, Ko S, Jung H, Kim B, Roh JH, Jun TJ, Kim YH. Time series forecasting of weight for diuretic dose adjustment using bidirectional long short-term memory. *Scientific Reports*. 2024 Jul 31;14(1):17723.

Under Review

- **Heejung Choi**, Yunha Kim, Jiahn Seo, Gaeun Kee, Soyoung Ko, HyoJe Jung, Byeolhee Kim, Boeun Choi, Ha Yeon Kim, Yoobin Park, Jihoon Yoon, Young-Hak Kim, Jungyo Suh, Tae Joon Jun. Task-specific performance of locally fine-tuned large language models for tabular clinical prediction: a controlled comparison with machine-learning models. 2026. *Scientific Reports* (Under Review).

Co-author

- Ahn I, Gwon H, Kim Y, Kang HJ, Seo H, **Choi H**, Cho HN, Kim M, Han J, Kee G, Park S, Jun TJ, Kim YH. PREP: a practical RNN-based estimated glomerular filtration rate prediction for inpatients with heart failure. *BMC Medical Informatics and Decision Making*. 2026 Aug 26.

- Seo J, Kim Y, **Choi H**, Kim M, Han J, Kee G, Ko S, Jung H, Kim B, Choi B, Park S. Protecting medical privacy with pretrained language models: named entity recognition-based de-identification in Korean unstructured electronic medical records. *npj Health Systems*. 2025 Sep 4;2(1):32.

- Kim M, Kim Y, Kang HJ, Seo H, **Choi H**, Han J, Kee G, Ko S, Jung H, Kim B, Choi B. Leveraging BERT for embedding ICD codes from large scale cardiovascular EMR data to understand patient diagnostic patterns. *BMC Medical Informatics and Decision Making*. 2025 Aug 11;25(1):300.

- Jung H, Kim Y, Seo J, **Choi H**, Kim M, Han J, Kee G, Ko S, Kim B, Choi B, Kim AR. Clinical assessment of fine-tuned open-source LLMs in cardiology: from progress notes to discharge summary. *Journal of Healthcare Informatics Research*. 2025 Dec;9(4):686-702.

- Han J, Kim Y, Kang HJ, Seo J, **Choi H**, Kim M, Kee G, Park S, Ko S, Jung H, Kim B. Predicting low density lipoprotein cholesterol target attainment using machine learning in patients with coronary artery disease receiving moderate-dose statin therapy. *Scientific Reports*. 2025 Feb 13;15(1):5346.

- Kim M, Kim Y, Kang HJ, Seo H, **Choi H**, Han J, Kee G, Park S, Ko S, Jung H, Kim B. Fine-tuning LLMs with medical data: can safety be ensured?. *NEJM AI*. 2025 Jan 1;2(1):AIcs2400390.

- Cho HN, Ahn I, Gwon H, Kang HJ, Kim Y, Seo H, **Choi H**, Kim M, Han J, Kee G, Park S. Explainable predictions of a machine learning model to forecast the postoperative length of stay for severe patients: machine learning model development and evaluation. *BMC Medical Informatics and Decision Making*. 2024 Nov 20;24(1):350.

- Cho HN, Jun TJ, Kim YH, Kang H, Ahn I, Gwon H, Kim Y, Seo J, **Choi H**, Kim M, Han J. Task-specific transformer-based language models in health care: scoping review. *JMIR Medical Informatics*. 2024 Nov 18;12:e49724.

- Kim Y, Kang H, Seo H, **Choi H**, Kim M, Han J, Kee G, Park S, Ko S, Jung H, Kim B. Development and transfer learning of self-attention model for major adverse cardiovascular events prediction across hospitals. *Scientific Reports*. 2024 Oct 8;14(1):23443.

- Seo H, Ahn I, Gwon H, Kang H, Kim Y, **Choi H**, Kim M, Han J, Kee G, Park S, Ko S. Forecasting hospital room and ward occupancy using static and dynamic information concurrently: retrospective single-center cohort study. *JMIR Medical Informatics*. 2024 Mar 21;12:e53400.

- Kee G, Kang HJ, Ahn I, Gwon H, Kim Y, Seo H, **Choi H**, Cho HN, Kim M, Han J, Park S. Are polypharmacy side effects predicted by public data still valid in real-world data?. *Heliyon*. 2024 Jan 30;10(2).

- Gwon H, Ahn I, Kim Y, Kang HJ, Seo H, **Choi H**, Cho HN, Kim M, Han J, Kee G, Park S. LDP-GAN: generative adversarial networks with local differential privacy for patient medical records synthesis. *Computers in Biology and Medicine*. 2024 Jan 1;168:107738.

- Seo H, Ahn I, Gwon H, Kang HJ, Kim Y, Cho HN, **Choi H**, Kim M, Han J, Kee G, Park S. Prediction of hospitalization and waiting time within 24 hours of emergency department patients with unstructured text data. *Health Care Management Science*. 2024 Mar;27(1):114-29.

- Cho HN, Ahn I, Gwon H, Kang HJ, Kim Y, Seo H, **Choi H**, Kim M, Han J, Kee G, Jun TJ. Heterogeneous graph construction and HinSAGE learning from electronic medical records. *Scientific Reports*. 2022 Dec 7;12(1):21152.
- Kim Y, Ahn I, Cho HN, Gwon H, Kang HJ, Seo H, **Choi H**, Kim KP, Jun TJ, Kim YH. RIDAB: electronic medical record-integrated real world data platform for predicting and summarizing interactions in biomedical research from heterogeneous data resources. *Computer Methods and Programs in Biomedicine*. 2022 Jun 1;221:106866.
- Ahn I, Gwon H, Kang H, Kim Y, Seo H, **Choi H**, Cho HN, Kim M, Jun TJ, Kim YH. Machine learning–based hospital discharge prediction for patients with cardiovascular diseases: development and usability study. *JMIR Medical Informatics*. 2021 Nov 17;9(11):e32662.
- Gwon H, Ahn I, Kim Y, Kang HJ, Seo H, Cho HN, **Choi H**, Jun TJ, Kim YH. Self–training with quantile errors for multivariate missing data imputation for regression problems in electronic medical records: algorithm development study. *JMIR Public Health and Surveillance*. 2021 Oct 13;7(10):e30824.

Patents

Method and Apparatus for Predicting Initial Titration Dose of Anticoagulants Using Machine Learning
Republic of Korea Patent No. 10-2873330 Registered Oct 2025 (Filed Sep 2022)

Professional Training

Bio-signal & EMR AI Specialist Training Program
Korea Human Resource Development Institute for Health and Welfare (KOHI) May – Nov 2022

- Completed coursework in programming and statistics
- Conducted a team-based capstone project applying AI models to EMR data

Bio Data Engineer Training Program
Korean Society for Biotechnology and Bioengineering; Ministry of Science and ICT Oct 2020 – Feb 2021

- Completed coursework in introductory AI and statistics
- Collected biomedical data

Genomic Data Analysis Specialist Basic Course I
Seoul National University Hospital; Ministry of Health and Welfare Aug – Sep 2020

- Completed coursework in next-generation sequencing (NGS) fundamentals
- Performed Linux-based NGS data analysis